CSC1042 Computer Network II

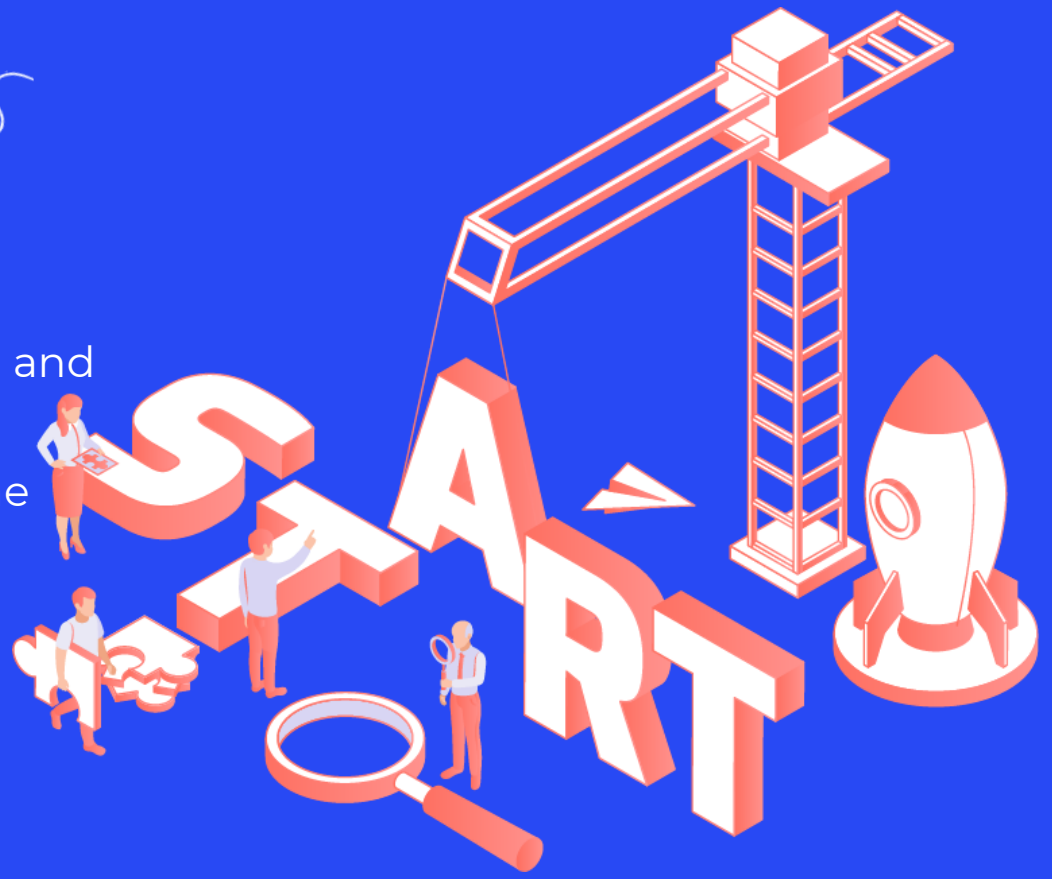
IP Routing

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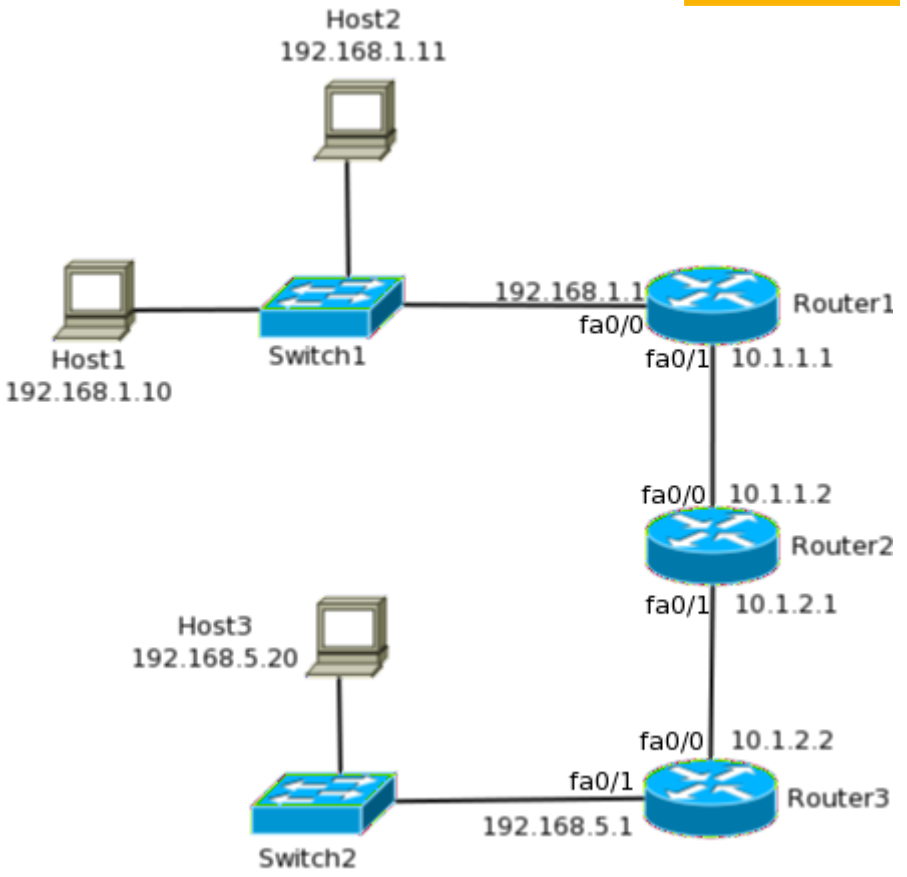
Learning outcomes

- Explain the concept of routing
- Compare the characteristics and use cases of static and dynamic routing
- Describe the fields of a routing table and explain the default route concept



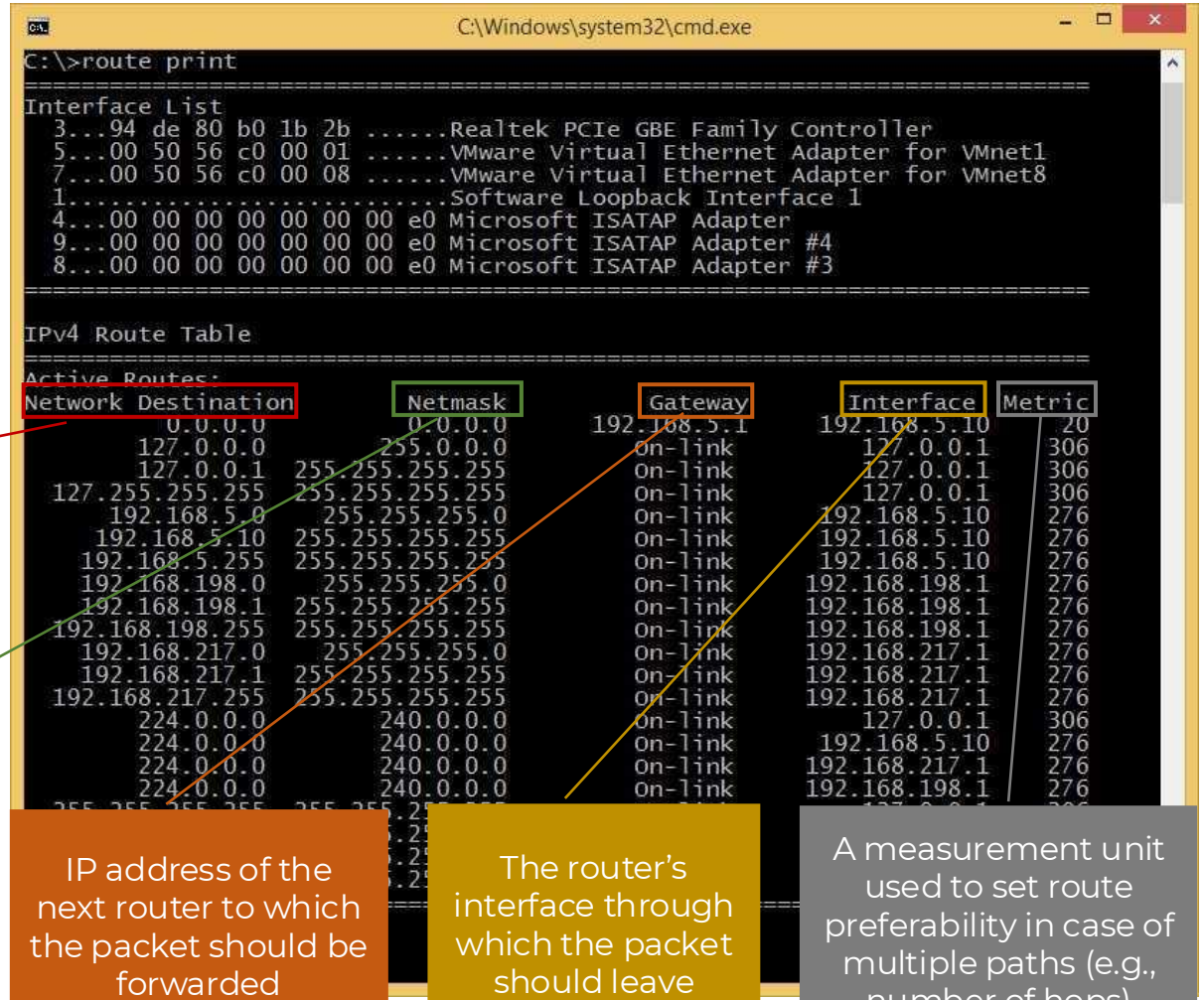
What's routing?

- Moving packets from one network to another
- It is performed by routers
- When receiving a packet, each router will read the destination IP address included in the packet's header to determine to which router it should be forwarded. We call it next hop
- Routers use routing tables to identify next hops
- The last router in the packet's path will deliver the packet to the destination



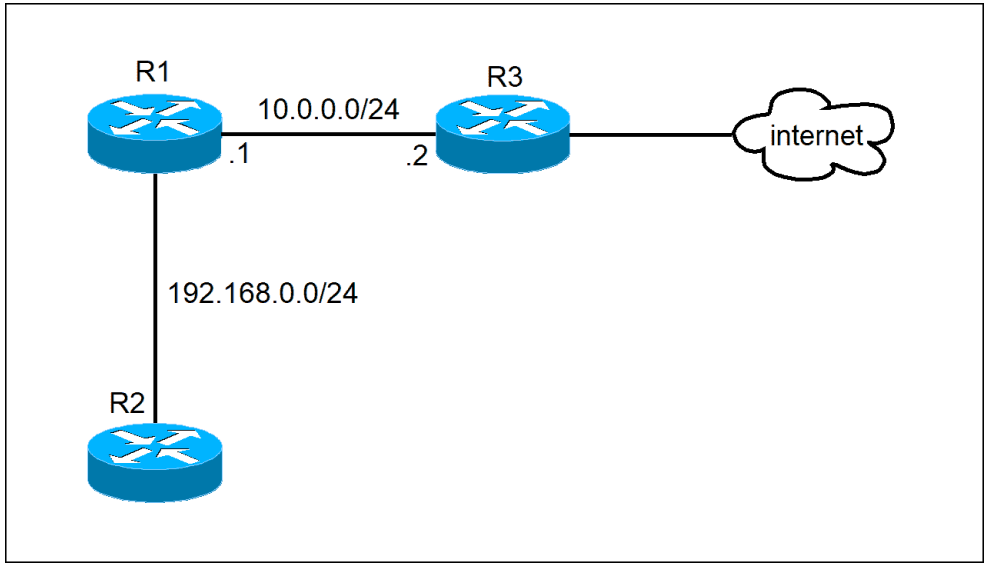
What's a routing table?

- A set of entries used to determine where data packets will be directed
- Each entry contains the information necessary to forward packets along the best path toward the destination



Default routes

- A default route defines where packets will be sent if no specific route for the destination network is listed in the routing table
- Without a default route, if R1 receives a packet with a destination IP address 200.10.3.4/24, the packet will be dropped
- With a configured default route, any packet with the destination network not listed in the routing table can be forwarded to R3
- For IPv4, the default route is 0.0.0.0/0



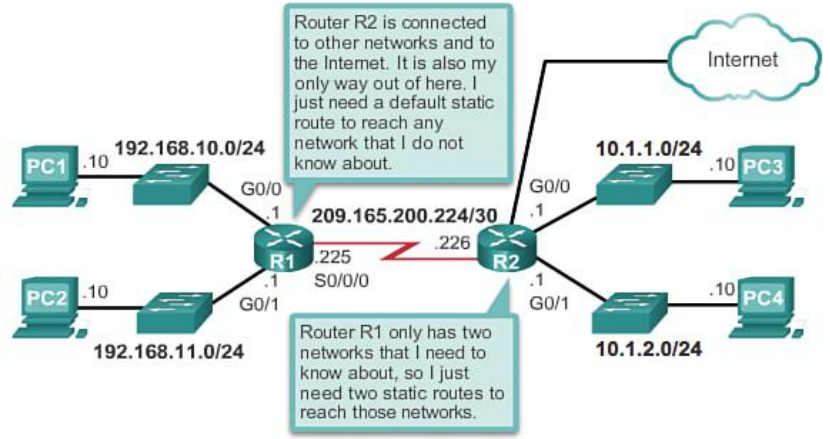
To configure the default route on R1, we can use the following command:

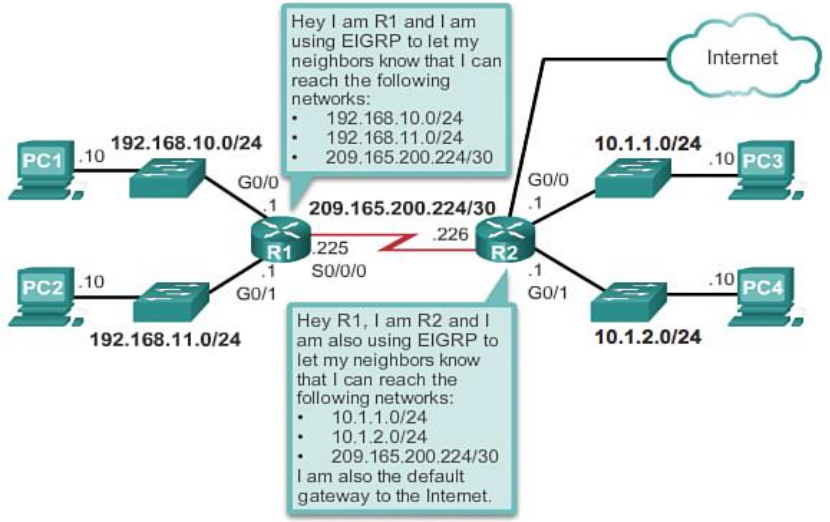
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R1(config)#ip route 0.0.0.0 0.0.0.0 10.0.0.2
```

This configures R1 to send all packets destined to the Internet to R3

How are routing tables populated?

- Routing tables can be populated in 2 way:
  - Static routing implies that routing tables are populated manually by the network admin
  - Dynamic routing indicates that entries in the routing tables are added automatically using a routing protocol





When to use static routing?

- Static routing is perfect for small size networks with few routes
- It provides some advantages, including:
  - Static routes are only advertised over the network, resulting in better security
  - Static routes use less bandwidth since routers do not exchange any routing messages with one another
  - The path a static route uses to send packets is known

When to not to use static routing?

- Static routing is not well suited for large networks as:
  - It requires complete knowledge of the whole network, which is extremely cumbersome in case of large networks
  - Administrator intervention is required when configuring and changing route information, which can be error-prone
  - Maintenance becomes cumbersome, especially in case of topology change
- Dynamic routing is used in large networks instead

Dynamic routing

- A technique of finding the best path through which packets can reach the destination
- Using routing protocols, routers exchange routing information to know how to reach remote networks
- They then compute the best path to these networks and add them to their routing tables
- Routers also determine alternative routes in case the best paths become less ideal or unavailable
- Dynamic routing uses two algorithms: distance vector and link state

Static routing vs. dynamic routing

	Dynamic Routing	Static Routing
Configuration complexity	Generally independent of the network size	Increases with the network size
Topology changes	Automatically adapts to topology changes	Administrator intervention required
Scaling	Suitable for simple and complex topologies	Suitable for simple topologies
Security	Less secure	More secure
Resource usage	Uses CPU, memory, link bandwidth	No extra resources needed
Predictability	Route depends on the current topology	Route to destination is always the same

THANK YOU